Trustworthy AI for Open RAN — Analysis Report

Audience: SMO / RIC operations

Generated:
[cci_a1sd_binary] xgboost — accuracy=1.000
Personal research project — not affiliated with any employer.

Contents

- Executive summary
- Dataset & model
- Base performance
- Explainability
- Robustness
- Fairness
- Privacy
- O-RAN console snapshot
- Agent trust evaluation
- Methodology & validity limits

Executive summary

Operational read-out for SMO / Non-RT RIC staff: model status, what to watch at the cell edge, and which thresholds should trigger an A1 policy review.

Key read-out

Metric	Value
Accuracy	0.9996
Elapsed sec	5.95
Overall	Not measured

Watch items

- Adversarial degradation costs 32.0 accuracy points at ε=0.3. Keep the Near-RT RIC rate-limit policy armed.
- Disparate impact ratio is 0.6897. Below 0.8 the A1 policy set should be rebalanced.
- Top attribution features to monitor: inter_arrival_std, response_time_mean, payload_size_std, payload_size_mean, inter_arrival_cv.

Recommended actions

- Membership-inference AUC is 0.4946. Above 0.65 the detector memorises training rows — retrain with a larger privacy budget or more data.

Dataset & model

Dataset

Metric	Value
Dataset	cci_a1sd_binary
Task	binary
Samples	24000
Features	10
Classes	Normal, Dos
Group attrs	policy_type_id, load_label, operation, policy_name
Class counts	Normal=12000; Dos=12000

The dataset describes A1 traffic: Normal vs DoS.

Model configuration

Metric	Value
Model	xgboost
Task	binary
Params	n_estimators=200; max_depth=6; learning_rate=0.1; subsample=0.9; colsample_bytree=0.9; random_state=42
Elapsed sec	5.95

Base performance

Headline metrics on the held-out test split.

Metric	Value
Accuracy	0.9996
F1 (macro)	0.9996
Roc auc	0.9998

Figures are rendered from the same run the panels display.



Class distribution

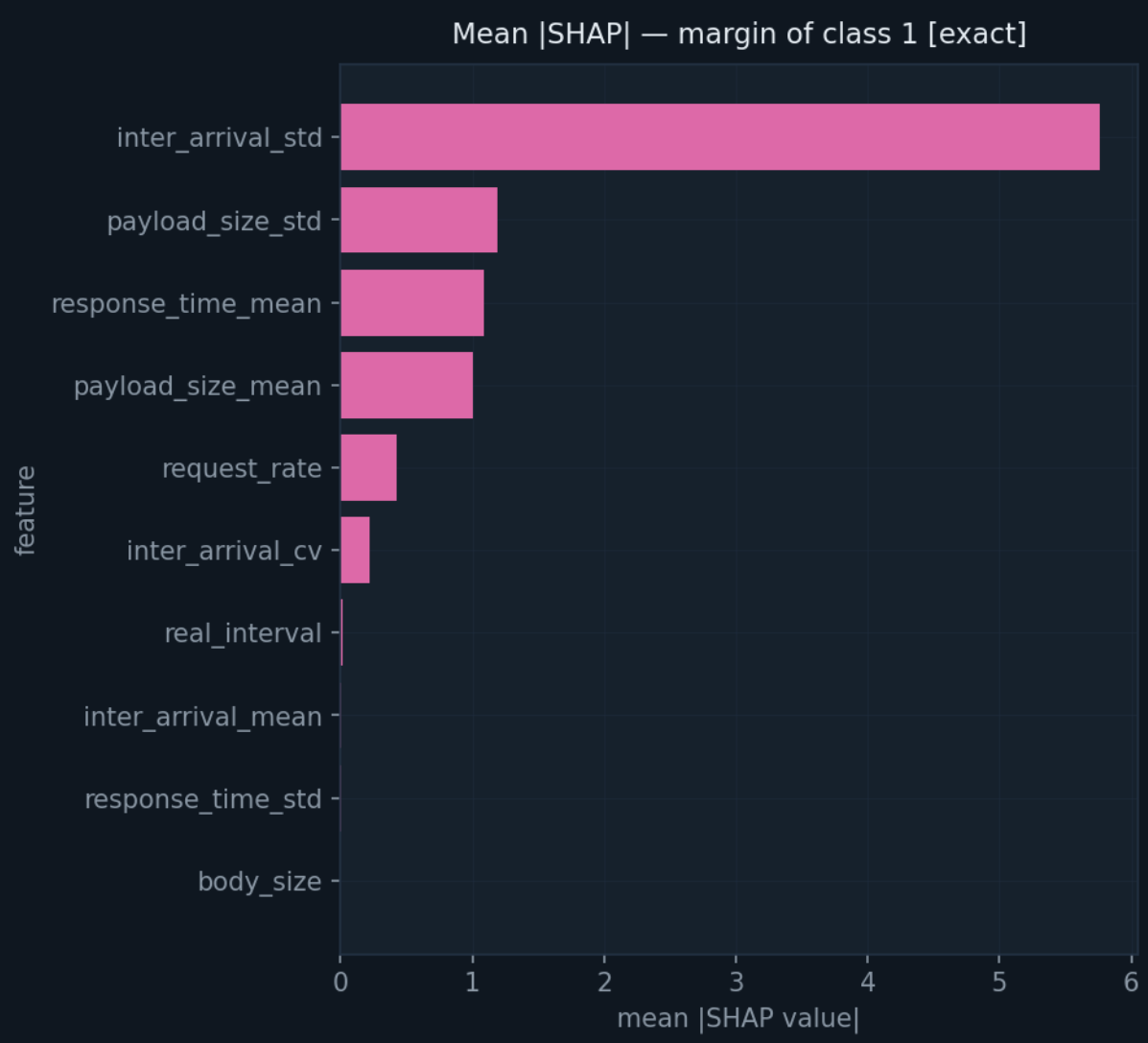


Feature histograms

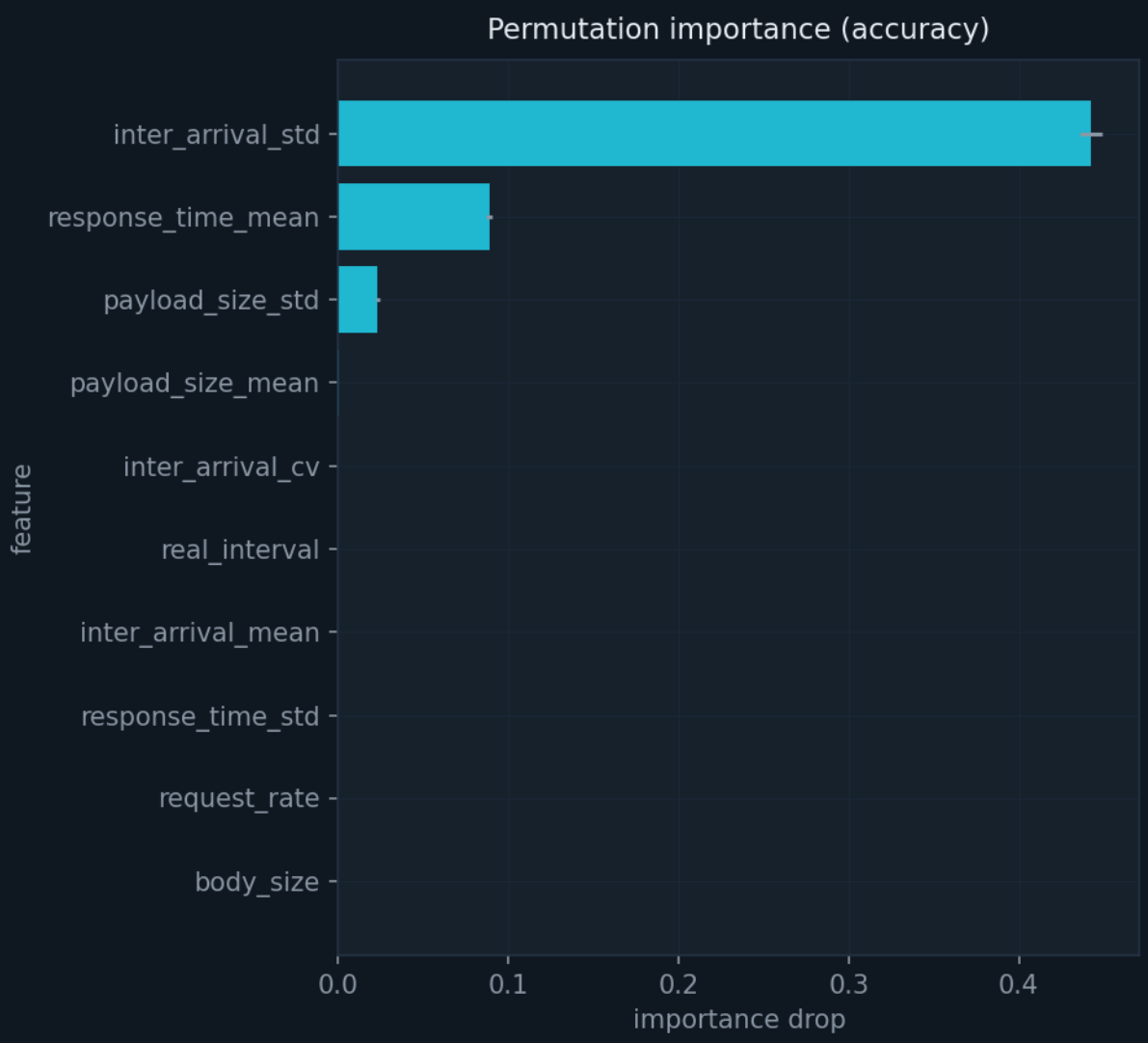
Explainability

Attribution tells the operator which A1 feature pushed a decision; use it when tuning policies and when disputing a false positive.

Metric	Value
Top features	inter_arrival_std, response_time_mean, payload_size_std, payload_size_mean, inter_arrival_cv
Target	margin of class 1
Shap top	inter_arrival_std, payload_size_std, response_time_mean, payload_size_mean, request_rate
Method	permutation + exact TreeSHAP (native pred_contribs) + LIME-style local + PDP/ICE
Shap method	exact
Shap efficiency max error	6.284e-06


Mean |SHAP| — margin of class 1 [exact]
inter_arrival_std
payload_size_std
response_time_mean
payload_size_mean
request_rate
inter_arrival_cv
real_interval
inter_arrival_mean
response_time_std
body_size
feature
0
1
2
3
4
5
6
mean |SHAP value|

Shap bar

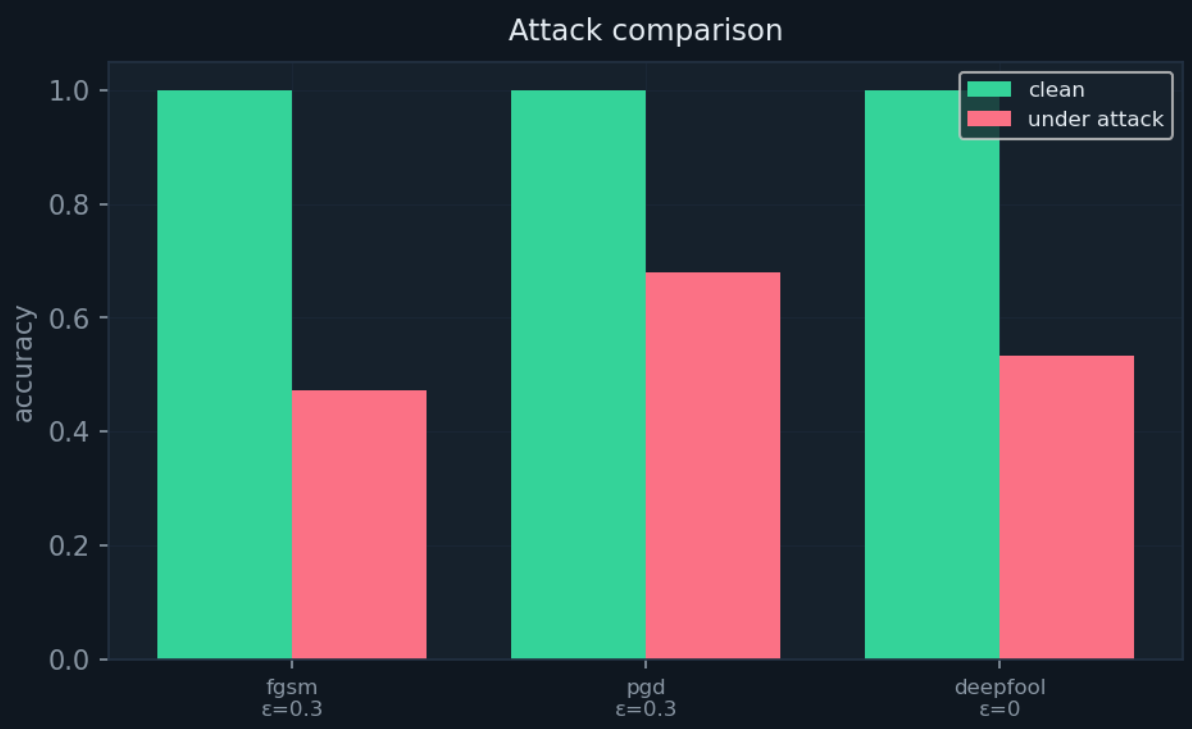


Permutation importance

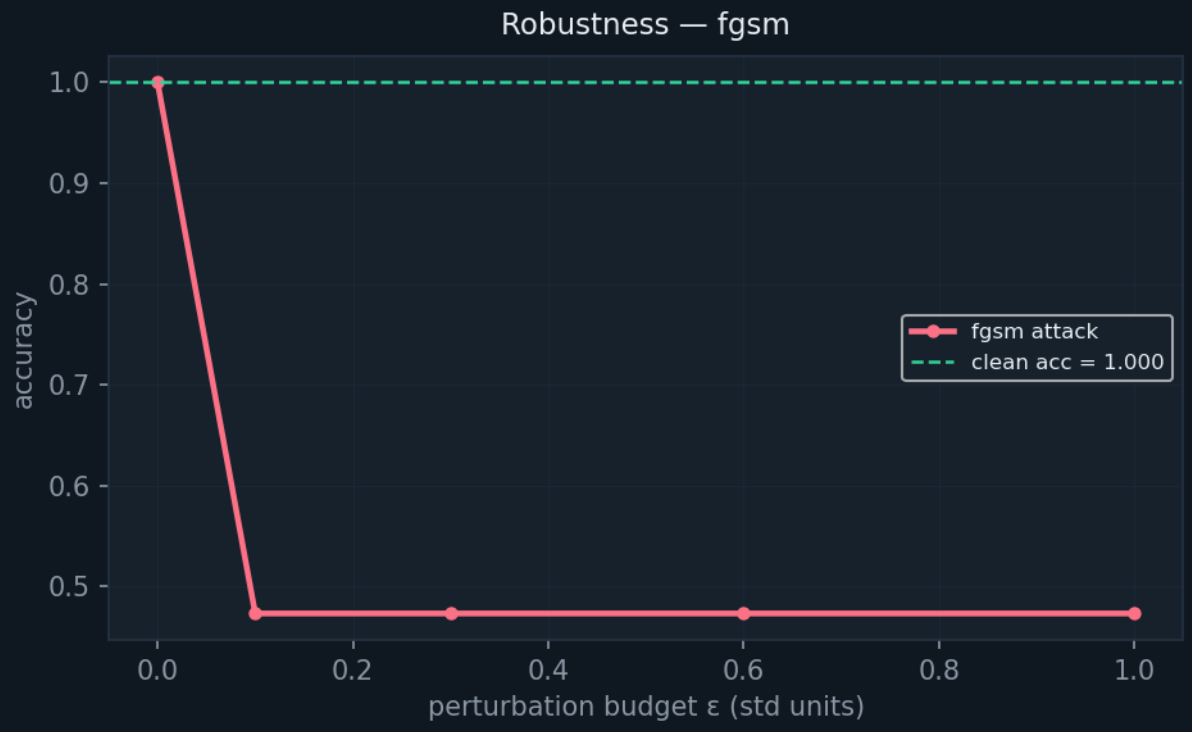
Robustness

Degradation under manipulated traffic — the realistic failure mode for an edge detector — plus the effect of adversarial training as a mitigation.

Metric	Value
Clean accuracy	1
Fgsm robust acc	0.4733
Pgd robust acc	0.68
Deepfool robust acc	0.5333
Pgd attack success rate	0.32
Adv trained robust acc@eps0.6	0.9667



Attack comparison



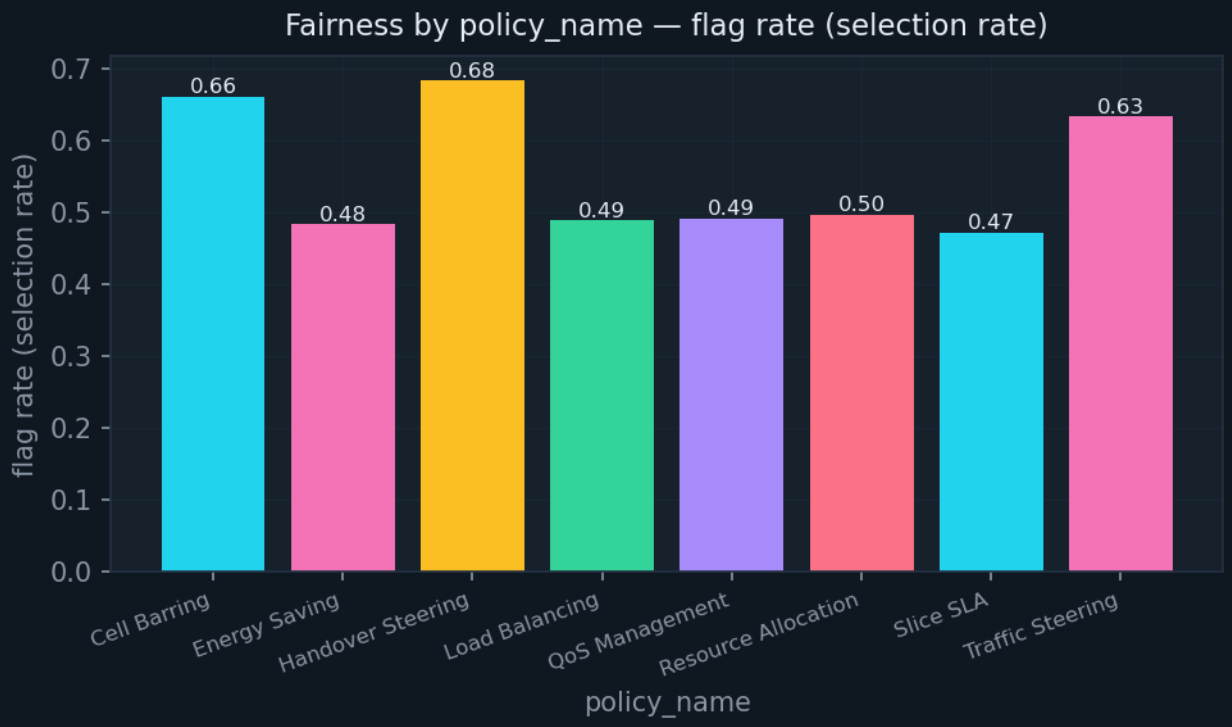
Robustness fgsm

Fairness

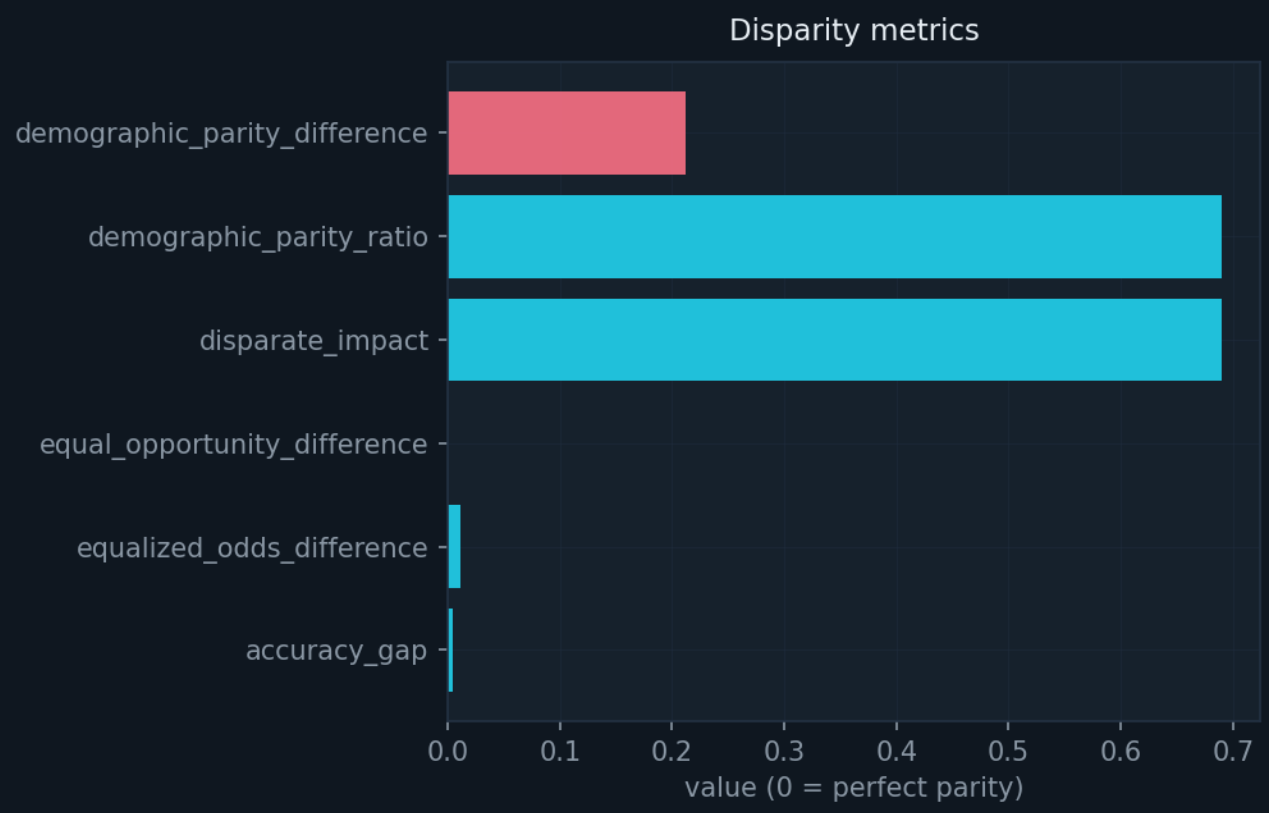
Group fairness by Policy name: whether the same alerting threshold behaves differently across policy types, agents or platforms.

Metric	Value
Group attr	policy_name
N groups	8

Metric	Value
Demographic parity difference	0.212
Demographic parity ratio	0.6897
Disparate impact	0.6897
Equal opportunity difference	0
Equalized odds difference	0.0116
Accuracy gap	0.0043



Fairness groups

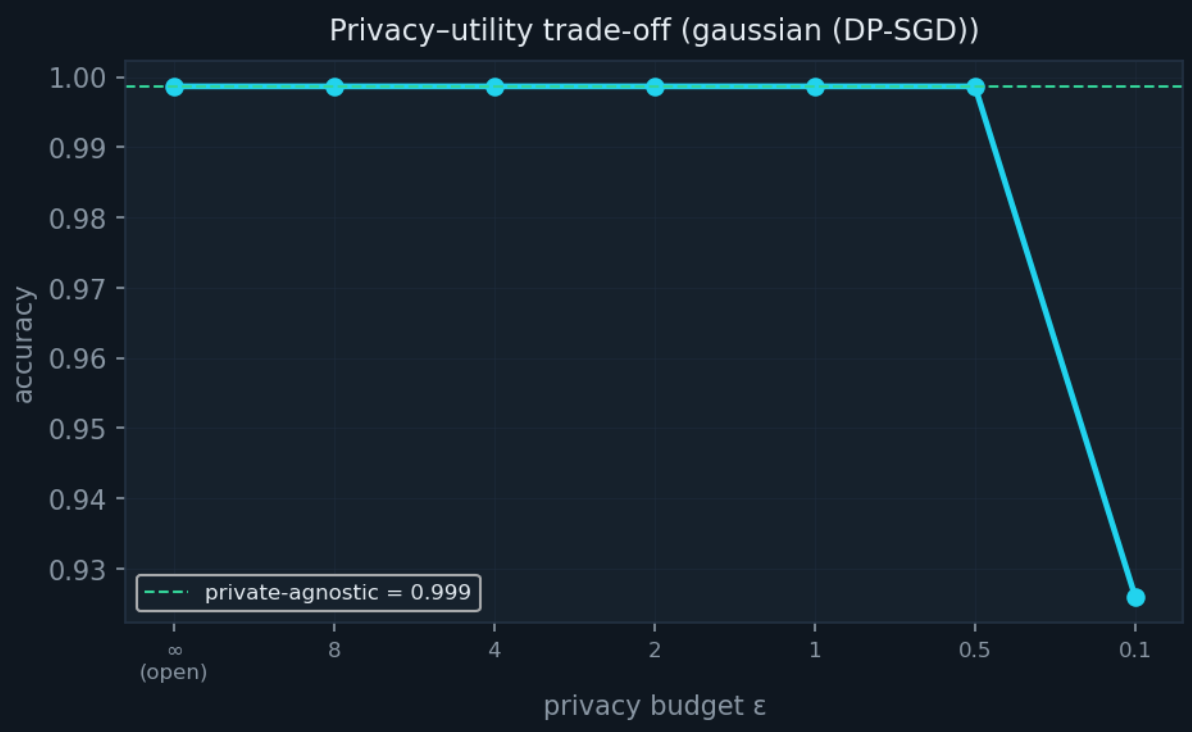


Fairness metrics

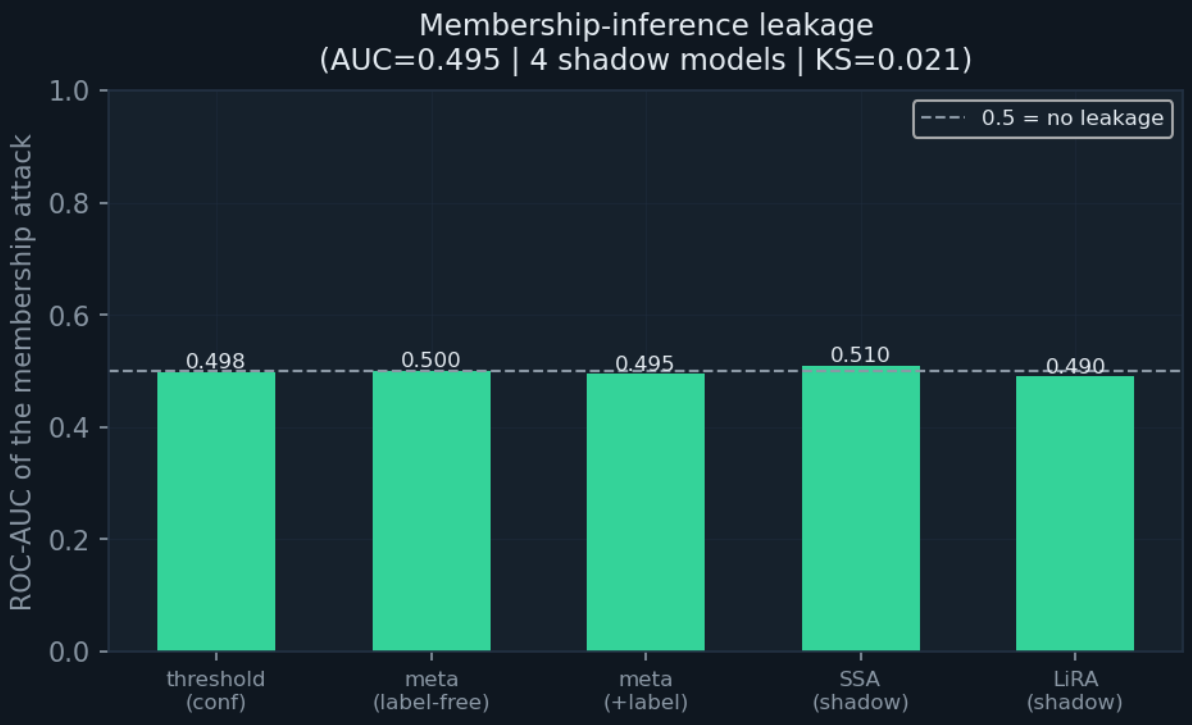
Privacy

Privacy budget and federation: how much utility is paid for differential privacy, and whether edge agents can learn without shipping raw traffic.

Metric	Value
Dp mechanism	gaussian (DP-SGD)
Utility open	0.9987
Utility eps1	0.9987
Utility eps0.1	0.926
Membership inference auc	0.4946
Mia shadow auc	0.51
Mia lira auc	0.4904
Mia conf only auc	0.5
Mia ks	0.02133
Mia tpr at 1pct fpr	0
Mia n shadows	4



DP curve



Membership inference

O-RAN console snapshot

No console session state was captured.

Agent trust evaluation

No agent evaluation has been run yet — run it in the Agent Eval panel to fill this section.

Methodology & validity limits

Reproducibility

```
python -m trustoran --dataset cci_a1sd_binary --model xgboost --pillars all --quick
python -m trustagent --quick
```

Validity limits

- Public research corpora — not production RAN telemetry.
- Attacks use differentiable surrogates, so robustness figures describe this attack family rather than an exhaustive adversary.
- Privacy accounting follows the moments accountant inside the local DP-SGD loop; no formal (ε,δ) certificate is issued for the shipped estimator.
- The fairness grouping is a modelling choice fixed per dataset in DATASET_PROFILES, not a property of the data itself.
- The agent environment is a sandbox with ACLs, budgets and confirmation gates; no real system or network element is touched.
- Mock-mode agent numbers validate the harness only; they are not a measurement of any real model.
- For normative wording, consult the O-RAN Alliance specifications directly.

This module is a personal research project that extends trustworthy-AI methodology to agent evaluation; it is not affiliated with any employer. The O-RAN scenarios and standards summaries are the author's own paraphrasing of public material.